ASSESSMENT TOOL 1 – Coding Challenge

CSA07-CN CO4 AT1

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Challenge Set 1: UDP Analysis

1.a.Locate the UDP header:

```
˃ Frame 1: 88 bytes on wire (704 bits)
˃ Ethernet II
˃ Internet Protocol Version 4
˅ User Datagram Protocol
    Source Port: 59198
    Destination Port: 53
    Length: 54
  ˃ Checksum: 0x37cb [correct] (matches partial checksum, not 0x647a, likely caused by "UDP checksum offload")
    Checksum Status: Good
    Stream index: 0
    Stream Packet Number: 1
  ˃ Timestamps
    UDP payload (46 bytes)
˃ Domain Name System (query)
```

b.Write down the Source Port and Destination Port

=>Source port: 59198

=>Destination port: 53

c.What is the Destination Port number and which well-known service does it represent?

=>Destination port number: 123 and It represents the **Network Time Protocol (NTP)** service, which is used for **time synchronization** over networks.

2.a.Find the UDP Length and Checksum fields.

UDP Length: `54 bytes`

UDP Checksum: `0x37ch`

b.Does this packet have any sequence number or acknowledgement number? What does this tell you about UDP?

- Sequence Number: No
- Acknowledgement Number: No

What this tells you about UDP:
UDP does not use sequence numbers or acknowledgement numbers. This means UDP is connectionless and does not provide reliable delivery, packet ordering, or retransmission of lost packets.

3.Opened NTP Sample:

```
˃ Frame 1: 90 bytes on wire (720 bits)
˃ Ethernet II
˃ Internet Protocol Version 4
˅ User Datagram Protocol
    Source Port: 123
    Destination Port: 123
    Length: 56
  ˃ Checksum: 0x70e1 [correct]
    Checksum Status: Good
    Stream index: 0
    Stream Packet Number: 1
  ˃ Timestamps
    UDP payload (48 bytes)
˃ Network Time Protocol (NTP Version 3)
```

a.Compare its UDP header with the DNS sample.

The UDP header in the **NTP sample** is similar to the UDP header in the **DNS sample**. Both contain the same four main fields: **Source Port, Destination Port, Length, and Checksum**. In the NTP sample, the source and destination ports are both **123**, while in the DNS sample, the source port is **59198** and the destination port is **53**. The NTP packet has a length of **56 bytes**, whereas the DNS packet has a length of **54 bytes**. Therefore, the UDP header structure is the same, but the port numbers, length, and checksum values can differ depending on the application.

b.**Are the header fields the same size?**

Yes. The UDP header has the same structure and field sizes for both DNS and NTP.

The standard UDP header is **8 bytes**:

- Source Port → **16 bits**
- Destination Port → **16 bits**
- Length → **16 bits**
- Checksum → **16 bits**

The **payload/data size changes**, but the UDP header remains the same.**c. Why is UDP suitable for both DNS and NTP?**

UDP is suitable because both DNS and NTP usually exchange **small amounts of data quickly**.

- **DNS:** needs fast request/response communication and does not require a connection to be established first.
- **NTP:** needs quick timing information with low overhead; waiting for a TCP connection would add unnecessary delay.

Challenge Set 2: TCP Analysis

4.

the following fields with their values:

Source Port: 21

Destination Port: 62014

Sequence Number: 0x836f4ffe

Acknowledgement Number: 0x04846eda

Flags (which flags are set?): PSH, ACK

5.a.What is the 4-tuple that uniquely identifies this TCP socket?

the **4-tuple** is:**(192.168.1.231, 21, 192.168.1.182, 62014)**

b.Explain how this 4-tuple helps in demultiplexing.

The **4-tuple** helps in demultiplexing by allowing the operating system to identify **which TCP connection should receive an incoming packet**. It uses the **source IP address, source port, destination IP address, and destination port** to uniquely identify the connection. When a packet arrives, TCP checks these four values and delivers the packet to the correct application or socket. This allows multiple TCP connections to exist at the same time without mixing their data.

c. Are the Destination Ports the same or different?How does the operating system know which application should receive each packet?(Demultiplexing)

The FTP sample uses Destination Port 62014 in your packet, while the MySQL sample will normally use Destination Port 3306 when connecting to a MySQL server. Therefore, the destination ports are different.

The operating system uses TCP port numbers along with IP addresses to perform demultiplexing. When a packet arrives, the OS checks the source and destination IP addresses and ports to identify the correct TCP connection and deliver the data to the appropriate application. Thus, packets port 21/FTP and port 3306/MySQL are directed to their respective applications.

Challenge Set 3: TCP vs UDP Comparison

7.

```
Transmission Control Protocol
  Source Port: 21
  Destination Port: 62014
  Stream index: 0
  Stream Packet Number: 1
  Conversation completeness: Incomplete (0)
  TCP Segment Len: 70
  Sequence Number: 0x836f4ffe
  Sequence Number (raw): 2205110270
  Next Sequence Number: 71
  Acknowledgment Number: 0x04846eda
  Acknowledgment number (raw): 75787994
  1000 .... = Header Length: 32 bytes (8)
  Flags: 0x018 (PSH, ACK)
  Window: 1448
  Calculated window size: 1448
  Window size scaling factor: -1 (unknown)
  Checksum: 0x9967 [correct]
  Checksum Status: Good
  Urgent Pointer: 0
  Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
  Timestamps
  SEQ/ACK analysis
  TCP payload (70 bytes)
```

.a.Which protocol has a larger header?

The FTP protocol (TCP) has a larger header than DNS (UDP). UDP has a fixed header size of 8 bytes, while TCP has a minimum header size of 20 bytes and can be larger when options are present. Your FTP sample shows a 32-byte TCP header.

b.Which protocol provides reliability features (sequence numbers, flags, window)?

TCP (FTP) provides reliability features such as Sequence Number, Acknowledgement Number, Flags (PSH, ACK), and Window Size. These fields help TCP provide reliable and ordered delivery.

c.Which protocol is connection-oriented and which is connectionless? Give evidence.

FTP using TCP is connection-oriented, because TCP establishes a connection before transferring data and uses sequence numbers, acknowledgements, and other control information.

DNS using UDP is connectionless, because UDP does not establish a connection before sending data and has no sequence numbers, acknowledgement numbers, or window field.

8.Point out any fields that show TCP's duty of providing reliability and flow control.

Sequence Number: `0x836f4ffe` — helps TCP keep data in the correct order.

Acknowledgement Number: `0x04846eda` — confirms the data received.

Flags: `PSH, ACK` — the ACK flag indicates acknowledgement.

Window: `1448` — controls how much data can be sent before an acknowledgement is required, providing flow control.

Checksum: `0x9967 [correct]` — helps detect errors in the TCP segment.

Challenge Set 4: Modern Transport

10.

- Frame 1: 115 bytes on wire (920 bits)
- Ethernet II
- Internet Protocol Version 6
- User Datagram Protocol
  - Source Port: 50280
  - Destination Port: 443
  - Length: 61
  - Checksum: 0xf6cf [correct]
  - Checksum Status: Good
  - Stream index: 0
  - Stream Packet Number: 1
  - Timestamps
  - UDP payload (53 bytes)
- QUIC IETF

Underlying Transport Protocol: UDP
QUIC runs on top of UDP rather than directly on TCP.

Destination Port: 443
QUIC commonly uses UDP port **443** for HTTPS traffic.

How QUIC differs from classic TCP:
Although QUIC uses UDP, it provides many features that TCP provides, such as **reliable delivery, acknowledgements, packet numbering, flow control, and congestion control**. QUIC also provides **built-in encryption (TLS)** and supports faster connection establishment and multiple streams within one connection.